Human pregnancy module validation report

Aciclovir simulations

2025-07-18 14:39:42

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Note

This report has been created by running the simulations. The results have been stored in folder ' D:/GitHub repositories/ModelReporitory/Pregnancy-Models/ValidationReports/Results/SimulationResults/2025-07-18 14-38 '.

1 Introduction

This report presents the validation of the human pregnancy module using Aciclovir simulations. The module is designed to simulate the pharmacokinetics of drugs during pregnancy, taking into account physiological changes that occur during this period.

2 Data

Data from the following studies has been used for model validation:

	Frenkel 1991 [1]	Kimberlin 1998 [2]	Leung 2009 [3]
Dose	400 mg	(I) 200 mg (II) 400 mg	400 mg
Frequency	3 times daily	3 times daily	3 times daily
Route of administration	Oral	Oral	Oral
N	15	20	27
Duration	Until delivery	Until delivery	Until delivery
Gestational age at start of treatment	38 weeks	36 weeks	36 weeks
Measurement	Maternal Plasma	Maternal Plasma	Maternal plasma, Umbilical cord blood

3 Results

Comparison of individual simulations with observed data are presented in the following figures.

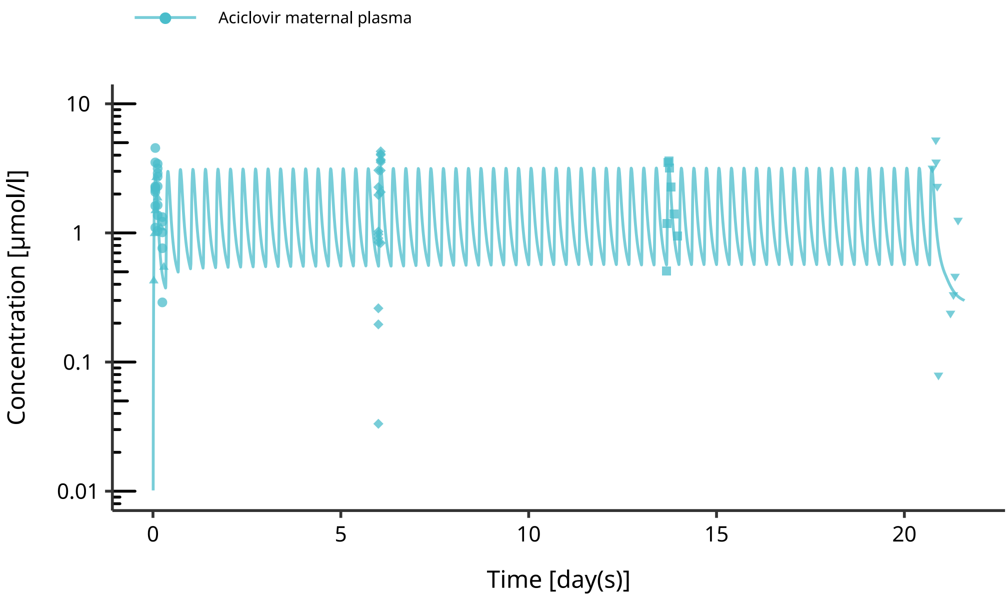

Figure 1: Simulation results of Acyclovir model in human pregnancy over multiple weeks. PK measurements present the following time points: (I) concentration after first dose, (II) steady state concentration (day 6 - 14) and (III) after delivery (>day 20).

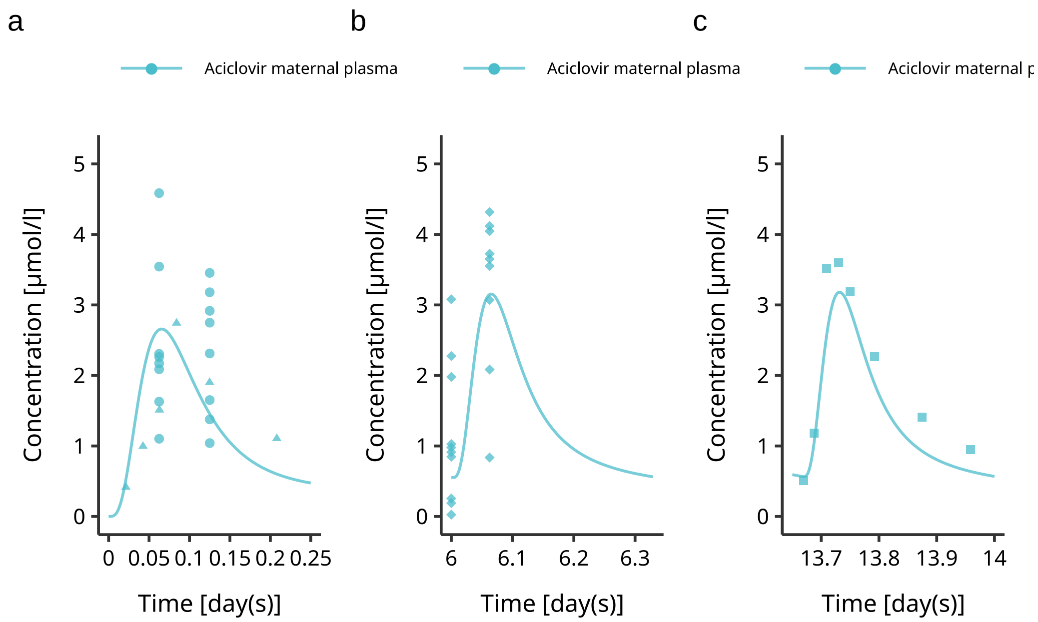

Figure 2: Simulation results of Acyclovir model in human pregnancy after first dose. (a) maternal plasma concentration after first dose. (b) and (c) maternal plasma concentration at steady state.

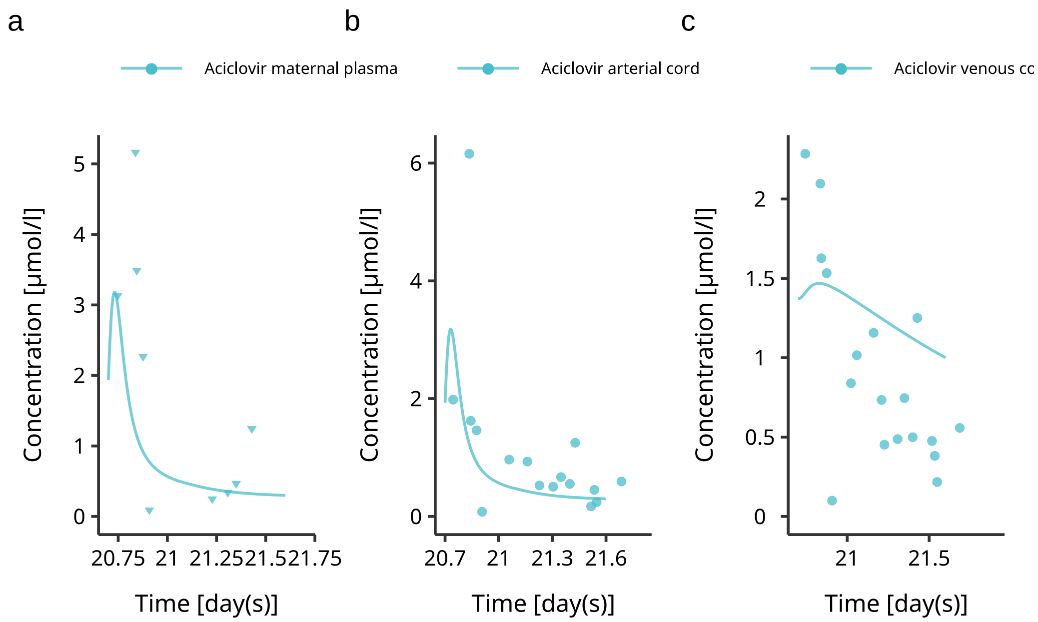

Figure 3: Simulation results of Acyclovir model in human pregnancy at delivery. (a) maternal plasma concentration after first dose. (b) and (c) maternal plasma concentration at steady state.

4 References

[1] Frenkel, Lisa M., et al. "Pharmacokinetics of acyclovir in the term human pregnancy and neonate." American journal of obstetrics and gynecology 164.2 (1991): 569-576.

[2] Kimberlin, Debora F, Stephen Weller, Richard J Whitley, William W Andrews, John C Hauth, Fred Lakeman, and Gerri Miller. 1998. "Pharmacokinetics of Oral Valacyclovir and Acyclovir in Late Pregnancy." American Journal of Obstetrics and Gynecology 179 (4): 846–51.

[3] Leung, Daniel T., Paul A. Henning, Emily C. Wagner, Audrey Blasig, Anna Wald, Stephen L. Sacks, Lawrence Corey, and Deborah M. Money. 2009. "Inadequacy of Plasma Acyclovir Levels at Delivery in Patients With Genital Herpes Receiving Oral Acyclovir Suppressive Therapy in Late Pregnancy." Journal of Obstetrics and Gynaecology Canada 31 (12): 1137–43.